



MERIT EDUCATION GROUP

What Is a Variable?

I can identify what a variable is and explain how it represents an unknown quantity.

Standard: 9.3.5.3

Standard: 9.3.5.7

Standard: 9.3.5.8

Day 1 · Unit 1 · Theme: Minnesota State Fair

1 BELL RINGER — WHITEBOARDS READY!

Write the expression for “a number increased by 6” and hold up your board.

Your response: _____

2 KEY VOCABULARY

Variable — A _____, usually a letter, represents an unknown or changing quantity.

Expression — A math phrase made of numbers, variables, and operation _____ (no equals sign).

Coefficient — The number _____ by a variable in a term (3 in $3n$). _____

Constant — A term that is a number on its own, with _____ variable. _____

Evaluate — To find the _____ of an expression by replacing variables with numbers.

3 DIRECT INSTRUCTION — TAKE NOTES (I DO)

Example: A number n is tripled, then increased by 4. Write an algebraic expression.

• Step 1 — The unknown number is represented by the variable _____. (n)

• Step 2 — “Tripled” means multiply by _____ $\rightarrow$ _____. ($3, 3n$)

• Step 3 — “Increased by 4” means add _____ $\rightarrow$ _____. ($4, 3n + 4$)

4 GUIDED PRACTICE — WORK TOGETHER (WE DO)

• x decreased by 9. The product of 6 and a number y .

5 CULTURE CONNECTION — MINNESOTA STORIES

Ojibwe 🗺️ (**Tribal Nations**) — Ojibwe families harvest wild rice (manoomin) together from canoes, measuring and counting each harvest’s pounds of grain.

Word problem: An Ojibwe elder counts $4x$ fish in one net and $3x$ fish in another net. Write an expression for the total number of fish. If $x = 7$ fish per net, how many fish are there total?

Language fact: Ojibwemowin (Ojibwe): bezhig, niizh, niswi, niiwin (1-4)

6 INDEPENDENT PRACTICE — YOU TRY!

- Twice a number increased by 3 equals 15. Write an equation and find the number.
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7 EXIT TICKET — SHOW WHAT YOU KNOW

Write an expression for “a number divided by 4, then increased by 7.”

Answer: _____

Show your work for full credit.